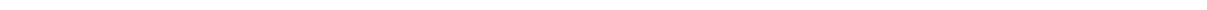


Advanced Metrological Characterization of Nanohenry Inductance and Capacitance in Metallized Film Capacitors

A Six-Method Comparative Analysis



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Auto-Balanced Bridge (ABB) Method

Theoretical Foundations & Mathematical Derivations

The Auto-Balanced Bridge (ABB) architecture is the primary instrumentation standard for high-precision impedance measurements from low frequencies up to approximately 120 MHz. The working mechanism relies on maintaining a virtual ground potential at the low terminal of the Device Under Test (DUT), thereby neutralising the effect of parasitic impedances to ground and isolating the complex impedance vector of the film capacitor.

The complex impedance of a real metallized film capacitor is modelled by the series equivalent circuit:

$$Z_x = R + j\left(\omega L - \frac{1}{\omega C}\right) \quad (1)$$

In the ABB configuration, a known AC voltage stimulus V_x is applied to the High-Force (H_{cur}) terminal of the DUT. The resulting current I_x flows through the capacitor to the Low-Force (L_{cur}) terminal. An active Operational Transimpedance Amplifier (OTA) detects any current flow at the L_{cur} node and dynamically adjusts its output voltage V_r through a known range-feedback resistor R_r to force the potential at the L_{cur} terminal to exactly zero volts. The nodal current-balancing equation at the virtual ground is:

$$I_x + I_r = 0 \implies \frac{V_x}{Z_x} + \frac{V_r}{R_r} = 0 \quad (2)$$

Solving for the complex impedance vector Z_x :

$$Z_x = -R_r \cdot \frac{V_x}{V_r} \quad (3)$$

Decomposing the voltage ratio into real and imaginary parts, $\frac{V_x}{V_r} = A + jB$, and substituting:

$$Z_x = -R_r(A + jB) = R_s + jX_s \quad (4)$$

where the apparent series resistance and reactance are:

$$R_s = -R_r A \quad (5)$$

$$X_s = -R_r B \quad (6)$$

At a single test frequency ω , a conventional instrument reports the apparent series capacitance C_s based on a simplified two-element capacitive model:

$$C_s = -\frac{1}{\omega X_s} = \frac{1}{\omega R_r B} \quad (7)$$

However, near or above the self-resonant frequency (SRF), the inductive reactance counteracts the capacitive reactance, leading to a systematic overestimation of C_s . To isolate

the true frequency-independent capacitance C and the nanohenry-range ESL (L), measurements must be performed at two distinct frequencies ω_1 and ω_2 within the capacitive regime:

$$X_{s1} = \omega_1 L - \frac{1}{\omega_1 C} \quad (8)$$

$$X_{s2} = \omega_2 L - \frac{1}{\omega_2 C} \quad (9)$$

Multiplying both equations by their respective angular frequencies:

$$\omega_1 X_{s1} = \omega_1^2 L - \frac{1}{C} \quad (10)$$

$$\omega_2 X_{s2} = \omega_2^2 L - \frac{1}{C} \quad (11)$$

Subtracting to eliminate the capacitive term yields the true ESL:

$$L = \frac{\omega_1 X_{s1} - \omega_2 X_{s2}}{\omega_1^2 - \omega_2^2} \quad (12)$$

Solving for the true capacitance C :

$$C = \frac{\omega_1^2 - \omega_2^2}{\omega_1 \omega_2^2 X_{s1} - \omega_1^2 \omega_2 X_{s2}} \quad (13)$$

Comprehensive Solved Numerical Example

The baseline DUT has the following true physical parameters:

$$\begin{aligned} C &= 2.919 \mu\text{F} = 2.919 \times 10^{-6} \text{ F} \\ L &= 20.90 \text{ nH} = 2.090 \times 10^{-8} \text{ H} \\ R &= 15.01 \text{ m}\Omega = 15.01 \times 10^{-3} \Omega \end{aligned}$$

Two test points are evaluated:

$$\begin{aligned} f_1 &= 100 \text{ kHz} & \Rightarrow \omega_1 &= 2\pi \times 10^5 \approx 628\,318.5307 \text{ rad/s} \\ f_2 &= 300 \text{ kHz} & \Rightarrow \omega_2 &= 2\pi \times 3 \times 10^5 \approx 1\,884\,955.5922 \text{ rad/s} \end{aligned}$$

At ω_1 :

$$\begin{aligned} X_{C1} &= -\frac{1}{\omega_1 C} = -\frac{1}{628318.5307 \times 2.919 \times 10^{-6}} \approx -0.545\,237\,2 \Omega \\ X_{L1} &= \omega_1 L = 628318.5307 \times 2.090 \times 10^{-8} \approx 0.013\,131\,9 \Omega \\ X_{s1} &= X_{L1} + X_{C1} = 0.0131319 - 0.5452372 = -0.532\,105\,3 \Omega \\ C_{s1} &= -\frac{1}{\omega_1 X_{s1}} = \frac{1}{628318.5307 \times 0.5321053} \approx 2.991\,101 \mu\text{F} \end{aligned} \quad (14)$$

At ω_2 :

$$\begin{aligned}
 X_{C2} &= -\frac{1}{\omega_2 C} \approx -0.181\,745\,7\,\Omega \\
 X_{L2} &= \omega_2 L \approx 0.039\,395\,6\,\Omega \\
 X_{s2} &= 0.0393956 - 0.1817457 = -0.142\,350\,1\,\Omega \\
 C_{s2} &= -\frac{1}{\omega_2 X_{s2}} \approx 3.725\,916\,\mu\text{F}
 \end{aligned} \tag{15}$$

Dual-frequency parameter reconstruction:

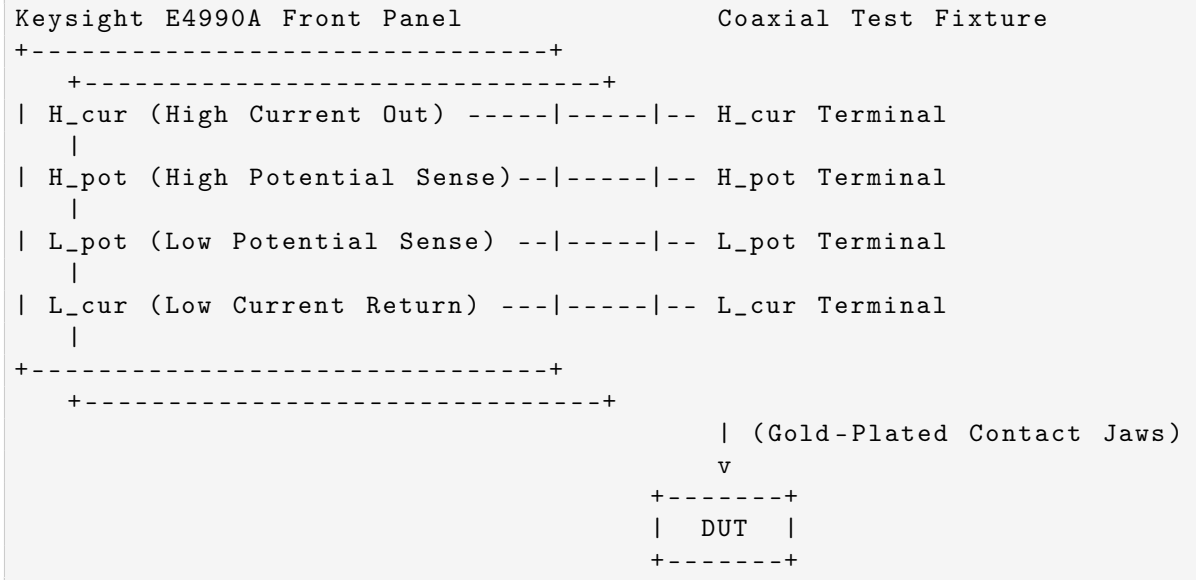
$$\begin{aligned}
 \text{Numerator} &= (628318.5307 \times -0.5321053) - (1884955.5922 \times -0.1423501) \\
 &= -334331.6136 - (-268323.4187) = -66008.1949 \\
 \text{Denominator} &= (628318.5307)^2 - (1884955.5922)^2 = 3.94784 \times 10^{11} - 3.55306 \times 10^{12} = -3.15827 \times 10^{12} \\
 L &= \frac{-66008.1949}{-3.15827 \times 10^{12}} \approx 2.090 \times 10^{-8} \text{ H} = 20.90 \text{ nH } \checkmark
 \end{aligned} \tag{16}$$

Reconstructing C :

$$\begin{aligned}
 \text{Numerator} &= -3.15827 \times 10^{12} \\
 \text{Denominator} &= -1.188044 \times 10^{18} - (-1.059293 \times 10^{17}) = -1.0821147 \times 10^{18} \\
 C &= \frac{-3.15827 \times 10^{12}}{-1.0821147 \times 10^{18}} \approx 2.918\,612 \times 10^{-6} \text{ F} \approx 2.919 \mu\text{F } \checkmark
 \end{aligned} \tag{17}$$

Practical Hardware Architecture & Bench Setup

The physical instrumentation consists of a Keysight E4990A Impedance Analyser interfaced through a Four-Terminal Pair (4TP) configuration.



Calibration follows a strict SOP:

1. **Open Compensation:** Record residual stray admittance $Y_{oc} = G_o + j\omega C_o$.
2. **Short Compensation:** Record residual series impedance $Z_{sc} = R_s + j\omega L_s$.
3. **Load Compensation:** Measure a precision $50\,\Omega$ coaxial resistor standard.

Analytical Accuracy & Uncertainty Budget

The phase angle at ω_1 :

$$\theta = \arctan\left(\frac{X_s}{R}\right) = \arctan\left(\frac{-0.5321053}{0.01501}\right) \approx -88.384^\circ \quad (18)$$

Table 1: ABB Method – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Phase Detector Resolution	Rectangular	$\pm 0.005^\circ$	$1.25 \times 10^{-10} \text{ H}/^\circ$	$\pm 0.063 \text{ nH}$
Residual Short Inductance	Normal ($k = 1$)	$\pm 0.150 \text{ nH}$	1.00	$\pm 0.150 \text{ nH}$
Range Resistor Drift	Triangular	$\pm 10 \text{ ppm}/^\circ\text{C}$	$4.20 \times 10^{-11} \text{ H/ppm}$	$\pm 0.042 \text{ nH}$
Guard-Loop Leakage	Rectangular	$\pm 0.05 \mu\text{A}$	$1.80 \times 10^{-4} \text{ H/A}$	$\pm 0.009 \text{ nH}$
Combined U_c	Normal ($k = 1$)	—	—	$\pm 0.168 \text{ nH}$

Long-Term Reliability & Stability Analysis

Measurement stability is highly sensitive to thermal drift of the internal range-feedback resistors R_r . A drift of $\pm 5^\circ\text{C}$ shifts the transimpedance amplifier gain, causing phase errors that translate into false inductive reactances. Lab temperature must be maintained at $23^\circ\text{C} \pm 1^\circ\text{C}$.

Calibration validity degrades through oxide formation on gold-plated contact tips, limiting the calibration lifespan to approximately **8 hours** under continuous use.

Formal Gauge R&R Framework

The measurement system variance is modelled using ANOVA decomposition:

$$\sigma_{\text{total}}^2 = \sigma_{\text{part}}^2 + \sigma_{\text{system}}^2 = \sigma_{\text{part}}^2 + \sigma_{\text{repeatability}}^2 + \sigma_{\text{reproducibility}}^2 \quad (19)$$

Repeatability (Equipment Variance, EV):

$$EV = 5.15 \cdot \sqrt{MSE} \quad (20)$$

Reproducibility (Appraiser Variance, AV):

$$AV = \sqrt{(5.15 \cdot \sigma_{\text{operator}})^2 - \frac{EV^2}{n \cdot r}} \quad (21)$$

Method-Specific Vulnerability: Manual clamping depth variance of $\pm 1.5 \text{ mm}$ introduces $\pm 1.5 \text{ nH}$ AV (7.1% relative error for the 20.90 nH baseline).

Mitigation Protocols:

- **Pneumatic Clamping Slides:** Constant contact force $6.5 \text{ N} \pm 0.1 \text{ N}$.
- **Ceramic Depth Stops:** Alumina ceramic stops limit insertion depth to $1.50 \text{ mm} \pm 0.05 \text{ mm}$.

Voltage Surge Discharge Method

Theoretical Foundations & Mathematical Derivations

The Voltage Surge Discharge Method is an active time-domain technique used to isolate *ESL* and *ESR* from the underdamped ringing current produced when a pre-charged capacitor discharges through a low-impedance switch. The discharge loop is modelled as a series *RLC* circuit governed by:

$$L \frac{d^2 i(t)}{dt^2} + R \frac{di(t)}{dt} + \frac{1}{C} i(t) = 0 \quad (22)$$

Dividing by L :

$$\frac{d^2 i(t)}{dt^2} + \frac{R}{L} \frac{di(t)}{dt} + \frac{1}{LC} i(t) = 0 \quad (23)$$

The characteristic roots are:

$$s_{1,2} = -\alpha \pm \sqrt{\alpha^2 - \omega_0^2} \quad (24)$$

where:

$$\alpha = \frac{R}{2L} \quad (\text{Neper frequency}) \quad (25)$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad (\text{undamped resonant frequency}) \quad (26)$$

For the underdamped condition ($\alpha < \omega_0$), the damped resonant frequency is:

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2} \quad (27)$$

With initial conditions $q(0) = CV_0$ and $i(0) = 0$:

$$i(t) = \frac{V_0}{\omega_d L} e^{-\alpha t} \sin(\omega_d t) \quad (28)$$

The damped period is measured from consecutive zero-crossings:

$$T_d = \frac{2\pi}{\omega_d} \implies \omega_d = \frac{2\pi}{T_d} \quad (29)$$

The logarithmic decrement from two consecutive positive peaks I_1, I_2 :

$$\delta = \ln\left(\frac{I_1}{I_2}\right) = \alpha T_d \implies \alpha = \frac{\delta}{T_d} \quad (30)$$

Then:

$$\omega_0 = \sqrt{\omega_d^2 + \alpha^2} \quad (31)$$

$$L_{\text{loop}} = \frac{1}{\omega_0^2 C} \quad (32)$$

$$R_{\text{loop}} = 2\alpha L_{\text{loop}} \quad (33)$$

After subtracting calibrated external parasitics:

$$L = L_{\text{loop}} - L_{\text{ext}} \quad (34)$$

$$R = R_{\text{loop}} - R_{\text{ext}} \quad (35)$$

Comprehensive Solved Numerical Example

True DUT parameters: $C = 2.919 \mu\text{F}$, $L = 20.90 \text{ nH}$, $R = 15.01 \text{ m}\Omega$.

Calibrated loop parasitics: $L_{\text{ext}} = 10.00 \text{ nH}$, $R_{\text{ext}} = 5.00 \text{ m}\Omega$.

Total loop: $L_{\text{loop}} = 30.90 \text{ nH}$, $R_{\text{loop}} = 20.01 \text{ m}\Omega$. Charge voltage $V_0 = 1000 \text{ V}$.

$$\alpha = \frac{0.02001}{2 \times 3.090 \times 10^{-8}} \approx 323\,786.4078/\text{s} \quad (36)$$

$$\omega_0 = \frac{1}{\sqrt{3.090 \times 10^{-8} \times 2.919 \times 10^{-6}}} \approx 3\,330\,031.5369 \text{ rad/s} \quad (37)$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2} \approx 3\,214\,198.5414 \text{ rad/s} \quad (38)$$

$$T_d = \frac{2\pi}{\omega_d} \approx 1.9548 \mu\text{s} \quad (39)$$

$$\delta = \alpha T_d \approx 0.632945 \quad (40)$$

First peak at:

$$t_{\text{peak}} = \frac{1}{\omega_d} \arctan\left(\frac{\omega_d}{\alpha}\right) \approx 457.46 \text{ ns} \quad (41)$$

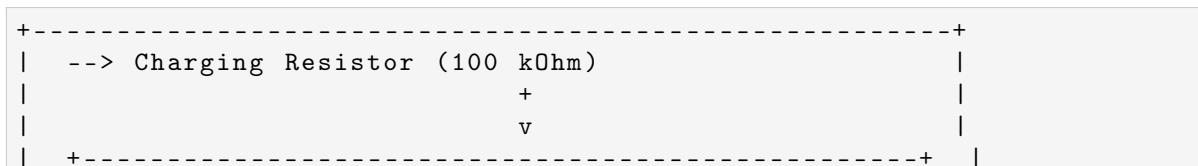
$$I_1 \approx 8638.64 \text{ A} \quad (42)$$

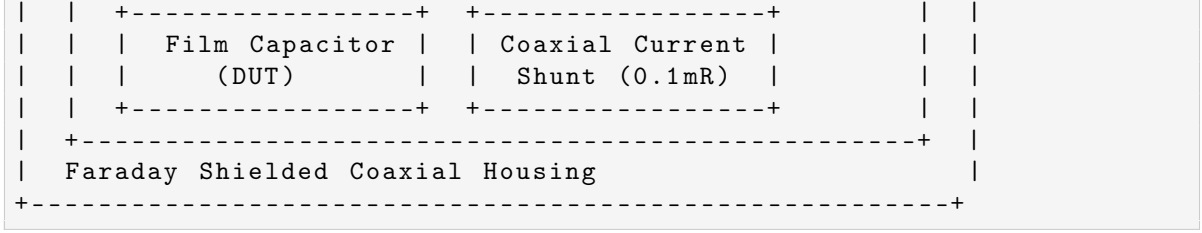
$$I_2 = I_1 e^{-\delta} \approx 4587.35 \text{ A} \quad (43)$$

Reconstruction yields $L_{\text{loop}} \approx 30.90$ nH, so:

$$L = 30.90 - 10.00 = 20.90 \text{ nH} \checkmark \quad (44)$$

Practical Hardware Architecture & Bench Setup





- **Switching Element:** Gas-filled spark gap, fall time < 10 ns.
- **Current Sensor:** Coaxial current viewing resistor, 1 GHz bandwidth, 0.1 m Ω .
- **Digitizer:** 2 GS/s, 10-bit real-time digitizer.

Analytical Accuracy & Uncertainty Budget

Table 2: Voltage Surge Discharge – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Digitizer Sampling Jitter	Rectangular	± 500 ps	3.16×10^{-2} nH/ns	± 0.016 nH
Spark Gap Dynamic Resistance	Normal ($k = 2$)	± 0.80 m Ω	1.50×10^{-9} H/ Ω	± 0.120 nH
Dummy Block Calibration Error	Triangular	± 0.05 nH	1.00	± 0.035 nH
Coaxial Shunt Thermal Drift	Normal ($k = 1$)	$\pm 0.2\%$	4.18×10^{-2} nH/%	± 0.084 nH
Combined U_c	Normal ($k = 1$)	—	—	± 0.151 nH

Formal Gauge R&R Framework

Vulnerabilities:

- Spark gap gas pressure variation ± 0.1 bar introduces $\Delta V = \pm 0.4$ nH.
- Inconsistent clamping torque varies contact resistance and loop inductance.

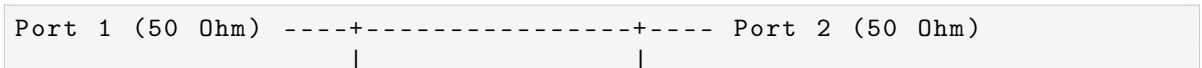
Mitigation:

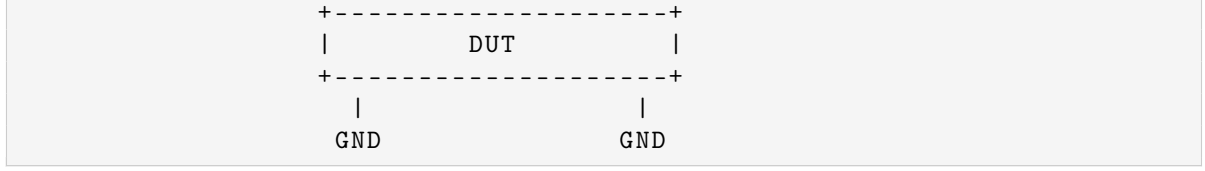
- Digital mass flow controllers at 2.20 bar ± 0.01 bar.
- Pneumatic torque wrenches at 4.5 N m ± 0.1 N m.

Vector Network Analyser (VNA) 2-Port Shunt-Thru Method

Theoretical Foundations & Mathematical Derivations

The 2-Port Shunt-Thru configuration is the metrological standard for ultra-low impedances ($\mu\Omega$ to Ω) from kHz to GHz. The DUT is connected in shunt between the two 50 Ω VNA ports:





The transmission coefficient is:

$$S_{21} = \frac{2Z_{\text{DUT}}}{2Z_{\text{DUT}} + Z_0} \quad (45)$$

Solving for Z_{DUT} :

$$Z_{\text{DUT}} = \frac{Z_0}{2} \cdot \frac{S_{21}}{1 - S_{21}} \quad (46)$$

For $Z_{\text{DUT}} \ll 25 \Omega$ ($S_{21} \ll 1$), the linear approximation holds:

$$Z_{\text{DUT}} \approx \frac{Z_0}{2} \cdot S_{21} = 25 \cdot S_{21} \quad (47)$$

With $S_{21} = \text{Re}(S_{21}) + j \text{Im}(S_{21})$, the exact formula after rationalising gives:

$$Z_{\text{DUT}} = 25 \cdot \frac{(\text{Re}(S_{21}) - |S_{21}|^2) + j \text{Im}(S_{21})}{(1 - \text{Re}(S_{21}))^2 + \text{Im}(S_{21})^2} \quad (48)$$

Parameter extraction over a wide sweep:

$$R(\omega) = \text{Re}(Z_{\text{DUT}}) \quad (49)$$

$$X(\omega) = \text{Im}(Z_{\text{DUT}}) = \omega L - \frac{1}{\omega C} \quad (50)$$

Above SRF ($f \gg f_{\text{SRF}}$): $L \approx \frac{\text{Im}(Z_{\text{DUT}})}{\omega}$.

Below SRF ($f \ll f_{\text{SRF}}$): $C \approx -\frac{1}{\omega \cdot \text{Im}(Z_{\text{DUT}})}$.

Comprehensive Solved Numerical Example

At $f = 1.0 \text{ MHz}$ ($\omega \approx 6.2832 \times 10^6 \text{ rad/s}$), above the 644 kHz SRF:

$$X_C = -\frac{1}{\omega C} \approx -0.054\,524 \Omega$$

$$X_L = \omega L \approx 0.131\,319 \Omega$$

$$X = 0.131319 - 0.054524 \approx 0.076\,795 \Omega$$

$$Z_{\text{DUT}} = 0.01501 + j0.076795 \Omega$$

Calculated S_{21} :

$$S_{21} \approx 6.09458 \times 10^{-4} + j3.06811 \times 10^{-3} \Rightarrow |S_{21}| \approx -50.09 \text{ dB} \quad (51)$$

Reconstructed impedance:

$$\operatorname{Re}(Z_{\text{DUT}}) \approx 15.01 \text{ m}\Omega \quad (52)$$

$$\operatorname{Im}(Z_{\text{DUT}}) \approx 76.79 \text{ m}\Omega \quad (53)$$

Using known C to correct:

$$L = \frac{0.07679 + \frac{1}{\omega C}}{\omega} = \frac{0.07679 + 0.054524}{6.2832 \times 10^6} \approx 2.090 \times 10^{-8} \text{ H} = 20.90 \text{ nH} \checkmark \quad (54)$$

Practical Hardware Architecture & Bench Setup

+-----+	+-----+	+-----+
VNA Port 1	J2102A	Coplanar
	Common Mode Xfmr	Microstrip
+-----+	+-----+	Probe Head
+-----+	+-----+	(Film Cap
VNA Port 2	J2113A	DUT Shunt)
	Semi-Float Amp	+-----+
+-----+	+-----+	

Ground-loop elimination requires:

- **J2102A** common-mode coaxial transformer on Port 1.
- **J2113A** differential amplifier on Port 2.
- Full 2-Port SOLT calibration at the capacitor terminal contacts.

Analytical Accuracy & Uncertainty Budget

IF bandwidth must be $\leq 10 \text{ Hz}$ to keep trace noise below 0.005 dB at the -50 dB transmission level.

Table 3: VNA 2-Port Shunt-Thru – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Dynamic Range Trace Noise	Normal ($k = 1$)	$\pm 0.005 \text{ dB}$	$1.20 \times 10^{-10} \text{ H/dB}$	$\pm 0.006 \text{ nH}$
Calibration Plane Phase Jitter	Normal ($k = 2$)	$\pm 0.02 \text{ rad}$	$2.50 \times 10^{-10} \text{ H/rad}$	$\pm 0.005 \text{ nH}$
Common-Mode Leakage	Rectangular	$\pm 0.05\%$	$1.80 \times 10^{-10} \text{ H}/\%$	$\pm 0.090 \text{ nH}$
Microstrip Probe Lead Drift	Triangular	$\pm 0.15 \text{ mm}$	$1.00 \times 10^{-9} \text{ H/mm}$	$\pm 0.087 \text{ nH}$
Combined U_c	Normal ($k = 1$)	—	—	$\pm 0.125 \text{ nH}$

Formal Gauge R&R Framework

Vulnerabilities:

- Cable flexing during loading alters phase delay (EV).
- Probe tip wear introduces parasitic shunt capacitance (EV).

Mitigation:

- Semi-rigid copper-jacketed cables clamped to an optical bench.
- Automated sub-micron 3-axis stage positioners with integrated pressure indicators.

Radio Frequency Current-Voltage (RF I-V) Method

Theoretical Foundations & Mathematical Derivations

The RF I-V method covers 1 MHz to 3 GHz using a passive coaxial transmission-line test head. The complex impedance is:

$$Z_x = \frac{V_x}{I_x} \quad (55)$$

With complex calibration scaling vectors $H_v(\omega)$, $H_i(\omega)$:

$$Z_x = \frac{H_v(\omega) \cdot V_v}{H_i(\omega) \cdot V_i} = Z_{\text{ref}}(\omega) \cdot \frac{V_v}{V_i} \quad (56)$$

Above SRF the inductance dominates. The correction formula for the true ESL :

$$L = \frac{X_s + \frac{1}{\omega C}}{\omega} = \frac{X_s}{\omega} + \underbrace{\frac{1}{\omega^2 C}}_{\text{correction}} \quad (57)$$

Comprehensive Solved Numerical Example

At $f = 10 \text{ MHz}$ ($\omega \approx 6.2832 \times 10^7 \text{ rad/s}$):

$$\begin{aligned} X_C &\approx -0.0054524 \Omega \\ X_L &\approx 1.3131857 \Omega \\ X_s &= 1.3131857 - 0.0054524 = 1.3077333 \Omega \\ Z_x &= 0.01501 + j1.3077333 \Omega \\ |Z_x| &\approx 1.3078194 \Omega, \quad \theta \approx 89.3424^\circ \end{aligned}$$

Raw (uncorrected) inductance:

$$L_{\text{raw}} = \frac{X_s}{\omega} = \frac{1.3077333}{6.2832 \times 10^7} \approx 20.813 \text{ nH} \quad (\text{error: } 0.41\%) \quad (58)$$

Applying the correction term:

$$\frac{1}{\omega^2 C} = \frac{1}{(6.2832 \times 10^7)^2 \times 2.919 \times 10^{-6}} \approx 8.678 \times 10^{-11} \text{ H} \approx 0.087 \text{ nH} \quad (59)$$

$$L = 20.813 + 0.087 = 20.90 \text{ nH} \checkmark \quad (60)$$

Analytical Accuracy & Uncertainty Budget

Table 4: RF I-V Method – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Residual Contact Inductance	Normal ($k = 1$)	± 50 pH	1.0	± 0.050 nH
Phase Detector Gain Error	Normal ($k = 2$)	$\pm 0.15\%$	3.13×10^{-2} nH/%	± 0.023 nH
Low-Loss Cap Cal Drift	Rectangular	± 0.02 pF	8.20×10^{-2} nH/pF	± 0.002 nH
Transformer Thermal Drift	Triangular	± 12 ppm/ $^{\circ}\text{C}$	4.50×10^{-11} H/ppm	± 0.010 nH
Combined U_c	Normal ($k = 1$)	—	—	± 0.056 nH

Formal Gauge R&R Framework

Mitigation Protocols:

- Micrometer clamping slides: lead alignment within ± 50 μm .
- Calibrated pre-loaded springs: constant contact force $8.0 \text{ N} \pm 0.2 \text{ N}$.

Resonant Cavity / Perturbation Method

Theoretical Foundations & Mathematical Derivations

When a low-loss capacitor is inserted into a high- Q coaxial cavity resonator, it perturbs the cavity boundary conditions, shifting the resonant frequency and Q factor.

The input impedance of a shorted coaxial transmission line of length h :

$$Z_{\text{in}} = jZ_c \tan(\beta h), \quad \beta = \frac{\omega}{c} \quad (61)$$

At cavity resonance, $Z_{\text{DUT}} + Z_{\text{in}} = 0$. Equating imaginary parts and noting that $\frac{1}{\omega C} \rightarrow 0$ at high RF frequencies:

$$\omega L + Z_c \tan\left(\frac{\omega h}{c}\right) = 0 \quad (62)$$

Solving for the true series inductance:

$$\boxed{L = -\frac{Z_c}{\omega} \tan\left(\frac{\omega h}{c}\right)} \quad (63)$$

The ESR is extracted from the quality factor shift:

$$\Delta\left(\frac{1}{Q}\right) = \frac{1}{Q_L} - \frac{1}{Q_0} = \frac{P_{\text{DUT}}}{\omega W_s} \implies R = K \cdot \Delta\left(\frac{1}{Q}\right) \quad (64)$$

Comprehensive Solved Numerical Example

Cavity parameters: $Z_c = 50 \Omega$, $h = 0.15 \text{ m}$, $Q_0 = 1200$, designed for $f_0 = 500 \text{ MHz}$.

The transcendental equation with $\theta = 5 \times 10^{-10} \omega$:

$$41.80 \theta + 50 \tan(\theta) = 0 \implies \tan(\theta) = -0.836 \theta \quad (65)$$

Root (fundamental quarter-wave mode): $\theta \approx 2.0933 \text{ rad}$.

Perturbed resonant frequency:

$$f = \frac{10^9 \times 2.0933}{\pi} \approx 666.32 \text{ MHz} \quad (66)$$

The 20.90 nH ESL shifts the cavity from 500.00 MHz to 666.32 MHz.

Reconstruction:

$$\omega = 2\pi \times 666.32 \times 10^6 \approx 4.1866 \times 10^9 \text{ rad/s} \quad (67)$$

$$\beta h \approx 2.0933 \text{ rad} \quad (68)$$

$$L = -\frac{50}{4.1866 \times 10^9} \tan(2.0933) = -1.1943 \times 10^{-8} \times (-1.7500) \approx 20.90 \text{ nH} \checkmark \quad (69)$$

Analytical Accuracy & Uncertainty Budget

Table 5: Resonant Cavity – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Cavity Length Expansion Δh	Normal ($k = 1$)	$\pm 5 \mu\text{m}$	$1.40 \times 10^{-7} \text{ H/m}$	$\pm 0.007 \text{ nH}$
NA Peak Search Resolution	Normal ($k = 2$)	$\pm 100 \text{ Hz}$	$3.10 \times 10^{-17} \text{ H/Hz}$	$\approx 0.000 \text{ nH}$
Z_c Machining Variance	Rectangular	$\pm 0.05 \Omega$	$4.20 \times 10^{-10} \text{ H}/\Omega$	$\pm 0.021 \text{ nH}$
Coupler Loading Perturbation	Triangular	$\pm 0.10\%$	$1.50 \times 10^{-2} \text{ nH}/\%$	$\pm 0.015 \text{ nH}$
Combined U_c	Normal ($k = 1$)	—	—	$\pm 0.027 \text{ nH}$

Formal Gauge R&R Framework

Mitigation Protocols:

- Micrometer depth stop: insertion depth $3.00 \text{ mm} \pm 0.01 \text{ mm}$.
- Pneumatic collet actuators: clamping force $15.0 \text{ N} \pm 0.1 \text{ N}$.

Time Domain Reflectometry / Transmission (TDR/TDT) Method

Theoretical Foundations & Mathematical Derivations

TDR injects an ultra-fast voltage step into a $50\ \Omega$ coaxial path shunt-loaded by the DUT. The Laplace-domain reflection coefficient, assuming $R \ll Z_0$:

$$\Gamma(s) \approx \frac{sL - Z_0}{sL + Z_0} \quad (70)$$

For a unit-step input $V_{\text{inc}}(s) = \frac{V_0}{s}$, the inverse Laplace transform yields:

$$v_{\text{meas}}(t) = 2V_0 e^{-\frac{Z_0}{L}t} \cdot u(t) \quad (71)$$

The inductance is extracted from the exponential time constant τ :

$$\boxed{\tau = \frac{L}{Z_0} \implies L = Z_0 \cdot \tau} \quad (72)$$

For non-ideal (finite rise-time) step excitation, the integration method is used:

$$L = 2Z_0 \int_{t_1}^{t_2} \rho(t) dt \quad (73)$$

Comprehensive Solved Numerical Example

With $V_0 = 1.0\ \text{V}$ and $Z_0 = 50\ \Omega$:

$$\tau = \frac{L}{Z_0} = \frac{20.90 \times 10^{-9}}{50} = 418.0\ \text{ps} \quad (74)$$

$$v_{\text{meas}}(t) = 2.0 \cdot e^{-t/418.0\ \text{ps}}\ \text{V} \quad (75)$$

The oscilloscope tracks the decay from $2.0\ \text{V}$ to $2.0/e \approx 0.735\ 76\ \text{V}$, giving $t_\tau = 418.0\ \text{ps}$:

$$L = 50 \times 418.0 \times 10^{-12} = 2.090 \times 10^{-8}\ \text{H} = 20.90\ \text{nH} \checkmark \quad (76)$$

Integration method verification:

$$\text{Area} = \frac{L}{2Z_0} = \frac{2.090 \times 10^{-8}}{100} = 209.0\ \text{ps} \quad (77)$$

$$L = 2Z_0 \cdot \text{Area} = 2 \times 50 \times 209.0 \times 10^{-12} = 20.90\ \text{nH} \checkmark \quad (78)$$

Analytical Accuracy & Uncertainty Budget

Table 6: TDR Method – Inductance Uncertainty Budget

Error Source	Distribution	Std. Uncertainty	Sensitivity Coeff.	L Contribution
Oscilloscope Timing Jitter	Normal ($k = 1$)	± 500 fs	1.20×10^{-1} nH/ps	± 0.060 nH
Step Amplitude Fluctuations	Normal ($k = 2$)	$\pm 0.50\%$	4.20×10^{-2} nH/%	± 0.011 nH
CPW Z_0 Machining Tolerance	Rectangular	$\pm 0.25 \Omega$	4.20×10^{-10} H/ Ω	± 0.105 nH
Time Gating Window Drift	Triangular	± 2.0 ps	5.00×10^{-2} nH/ps	± 0.100 nH
Combined U_c	Normal ($k = 1$)	—	—	± 0.158 nH

Formal Gauge R&R Framework

Mitigation Protocols:

- Calibrated SMA torque wrenches: $0.90 \text{ N m} \pm 0.05 \text{ N m}$.
- Vacuum-assisted pick-and-place arms: lead alignment within $\pm 10 \mu\text{m}$.

Metrological Comparison Matrix

Table 7: Side-by-side performance of all six metrological methodologies

Method	Impedance Range	Frequency Range	Primary Failure Mode	GRR Risk
Auto-Balanced Bridge (ABB)	$10 \text{ m}\Omega - 100 \text{ M}\Omega$	$20 \text{ Hz} - 120 \text{ MHz}$	Op-amp loop gain rolloff at high frequencies causes anomalous negative C or ESR	Medium
Voltage Surge Discharge	$1 \text{ m}\Omega - 10 \Omega$	Time-domain (100 kHz to 5000 kHz equiv.)	Spark gap arc non-linearity and digitiser jitter degrade peak fitting for ESR	High
VNA 2-Port Shunt-Thru	$20 \mu\Omega - 10 \Omega$	$100 \text{ kHz} - 3 \text{ GHz}$	Coaxial ground loops bypass DUT, overestimating ESR	Low

(continued)

Method	Impedance Range	Frequency Range	Primary Failure Mode	GRR Risk
RF Current-Voltage (RF I-V)	100 m Ω – 50 k Ω	1 MHz – 3 GHz	Residual contact inductance in test-head adapter obscures true ESL	Medium
Resonant Cavity / Perturbation	1 m Ω – 100 m Ω	100 MHz – 1 GHz	Humidity and thermal drift shift baseline resonant frequency	Medium
Time Domain Reflectometry (TDR)	1 m Ω – 500 Ω	Transient step (10 MHz – 20 GHz equiv.)	Finite step rise time blurs inductive transition peak; gating window errors	Low